Medical Image Processing

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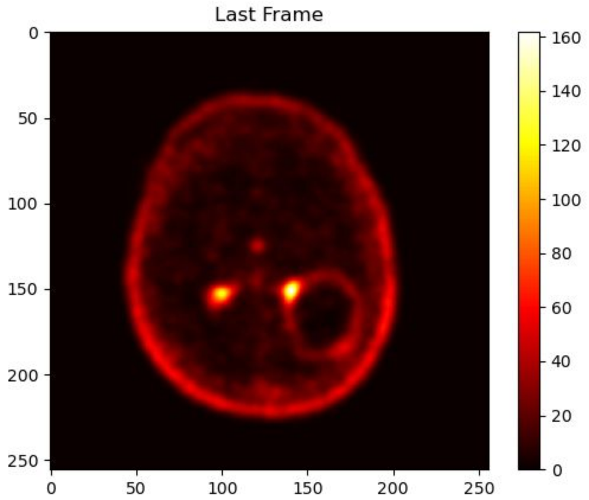
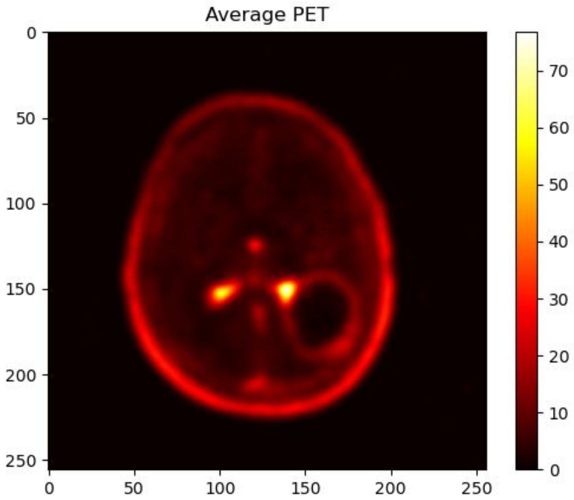
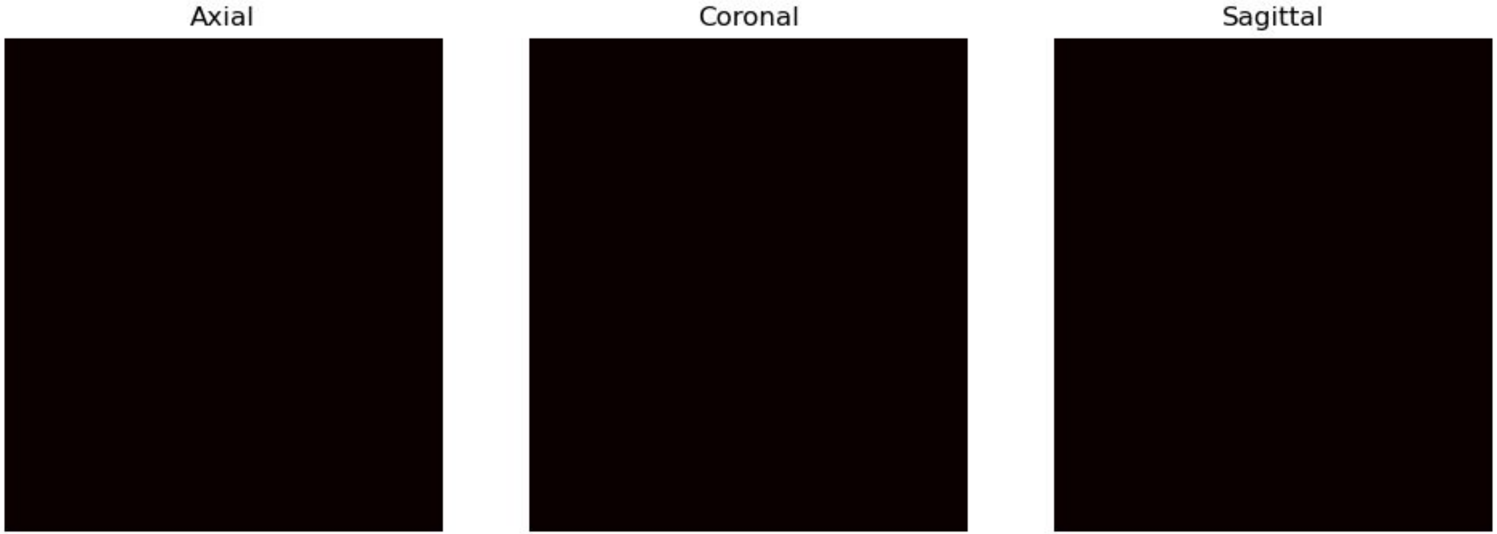
DICOM Loading and Visualization

- 3D Slicer
- Pydicom





DICOM Loading and Visualization

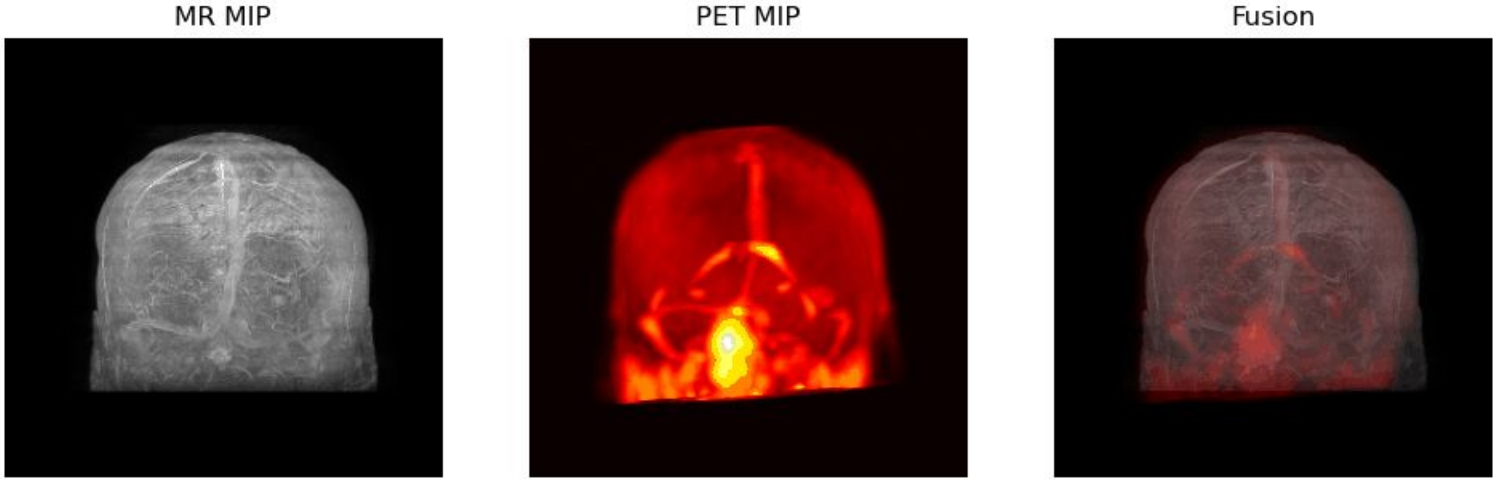


DICOM Loading and Visualization

```
ds_mr = pydicom.dcmread("./FORISI/15252129_s1_AX_3D_T1__C_FSPGR_FORISI260916")
mr_frames = int(ds_mr.get((0x0028, 0x0008), None).value)
mr_slices = ds_mr.get((0x0054, 0x0081), None).value
mr_rows = ds_mr.Rows
mr_cols = ds_mr.Columns
mr_slice_thickness = ds_mr.SliceThickness
mr_pixel_spacing = ds_mr.PixelSpacing
ds_mr = ds_mr.pixel_array.reshape(mr_frames // mr_slices, mr_slices, mr_rows, mr_cols)[-1, ::-1, :, :]
ds_mr = resample(ds_mr, (mr_slice_thickness, float(mr_pixel_spacing[0]), float(mr_pixel_spacing[1])))
ds_mr = resize_to_cube(ds_mr)
ds_mr = (normalize(ds_mr) / 255).astype(np.float32)
```

3D Rigid Coregistration

- SimpleITK



3D Rigid Coregistration

```
fixed = sitk.GetImageFromArray(ds_mr.astype(np.float32))

moving = sitk.GetImageFromArray(ds_pet.astype(np.float32))
```

3D Rigid Coregistration

```
initial_transform = sitk.CenteredTransformInitializer(
    fixed,
    moving,
    sitk.Euler3DTransform(),
    sitk.CenteredTransformInitializerFilter.GEOMETRY
)
```

3D Rigid Coregistration

```
reg = sitk.ImageRegistrationMethod()
```

3D Rigid Coregistration

```
reg.SetMetricAsMattesMutualInformation(numberOfHistogramBins=50)


reg.SetOptimizerAsRegularStepGradientDescent(
    learningRate=1.0,
    minStep=1e-4,
    numberOfIterations=1000,
    relaxationFactor=0.5
)
```

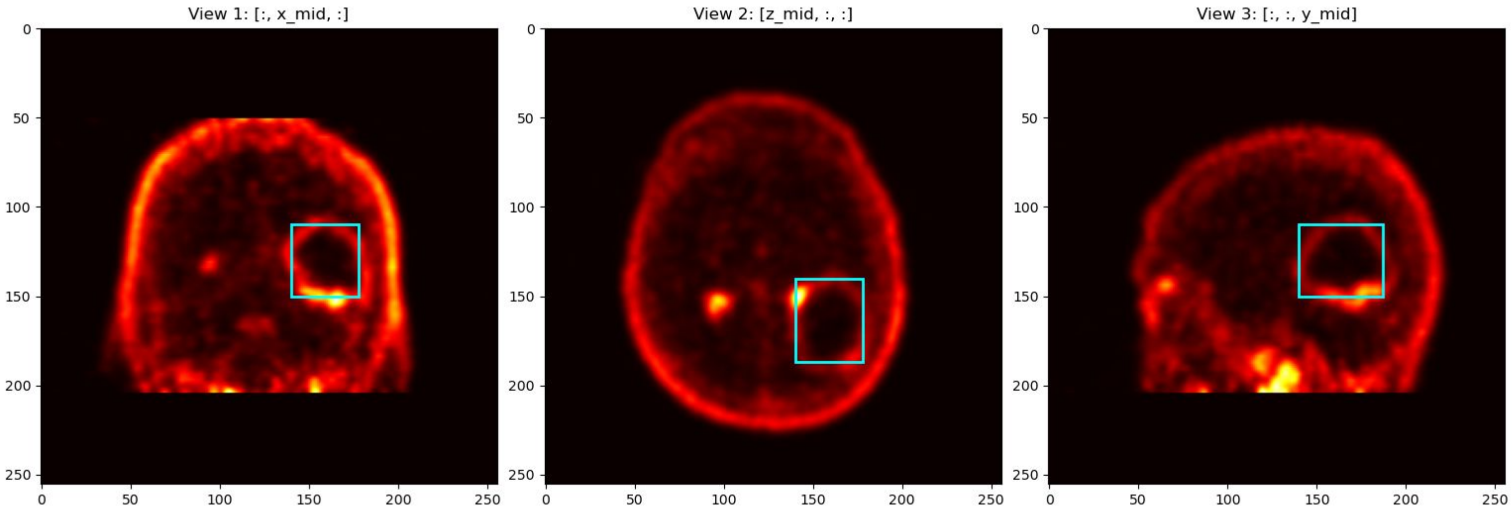
3D Rigid Coregistration

```
reg.SetInterpolator(sitk.sitkLinear)
```

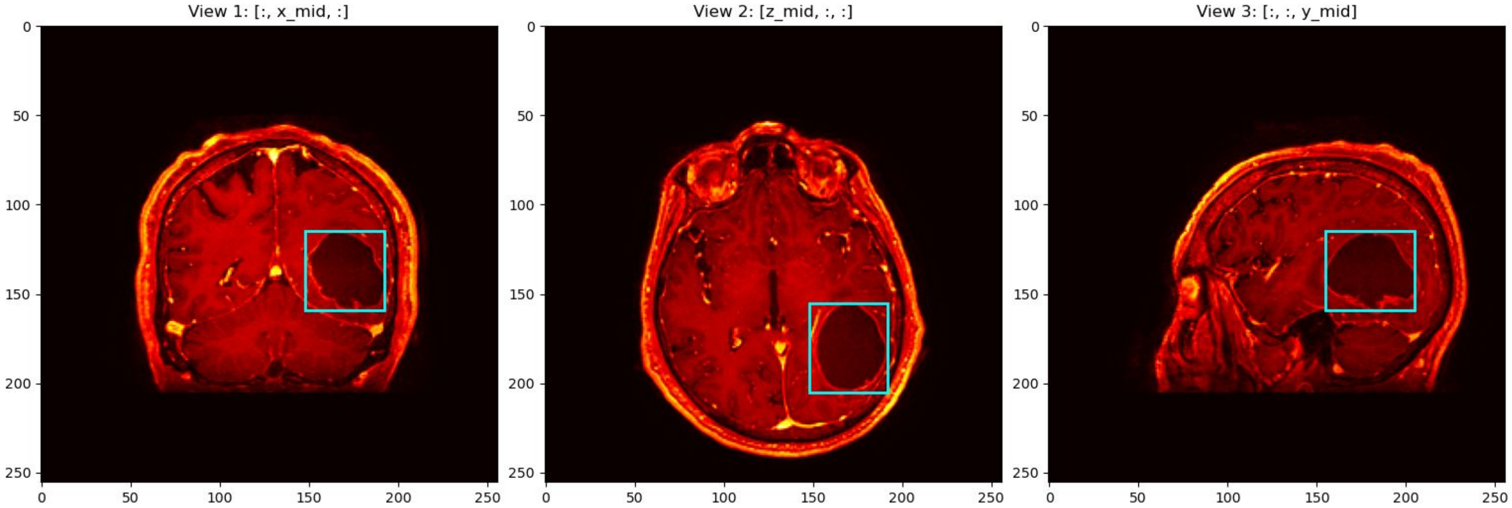
3D Image Segmentation

- Segment Anything Model (SAM)
- Vision Transformer backbone (ViT-H)

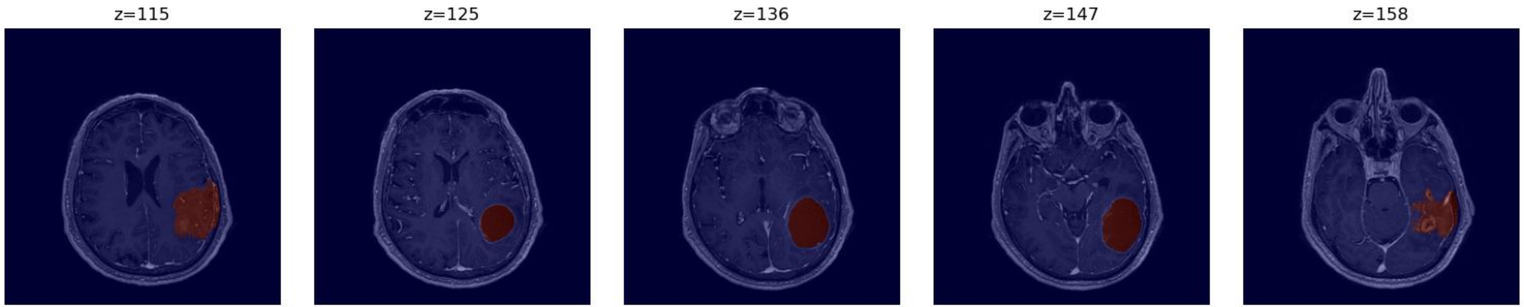
3D Image Segmentation



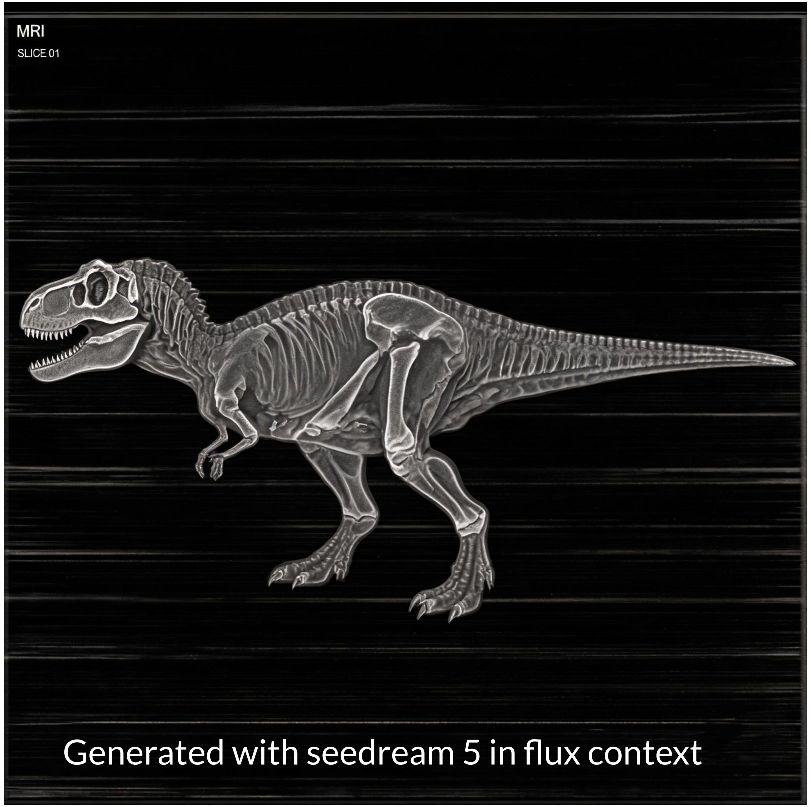
3D Image Segmentation



3D Image Segmentation



Thank You!



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